

China’s Import Engine - Case Study

China's Import	Introduction	Project overview	Data Info	Top imports
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CHINA'S IMPORT
ENGINE

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INTRODUCTION

China is the largest exporter and the second largest importer across the world. It relies on global trade to support its industrial production, energy supply, and consumer demand. This project analyzes China's import patterns by examining product categories, supplier countries, and transportation modes. The goal is to understand trade dependencies, logistics efficiency, and potential supply chain risks from both policy and logistics perspectives.



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PROJECT OVERVIEW & OBJECTIVES

The analysis focuses on understanding import dependencies, and overall trade balances to optimize trade policies and logistics efficiency.

- Identifying top commodities, destination markets, and transport efficiencies.
- Evaluating major suppliers, product dependencies, and cost structures



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DATA INFO:

ROW: 2,473,177
COLUMN: 19

DATA CLEANING PROCESS:

- CHANGE DATA TYPE
YEAR - DATE
AREA - GEOGRAPHICAL ROLE
- PERFORM CALCULATED COLUMNS FOR MEASURES;

Column: Description
Year: Transaction year

HS :Harmonized System (HS) category for traded goods.

Operation: Import or Export

Area: Country name for trade partner. (Exports = destination, Imports = source).

FobValue: Free on Board (FOB) value (in euros) representing the product's value at the trade point.

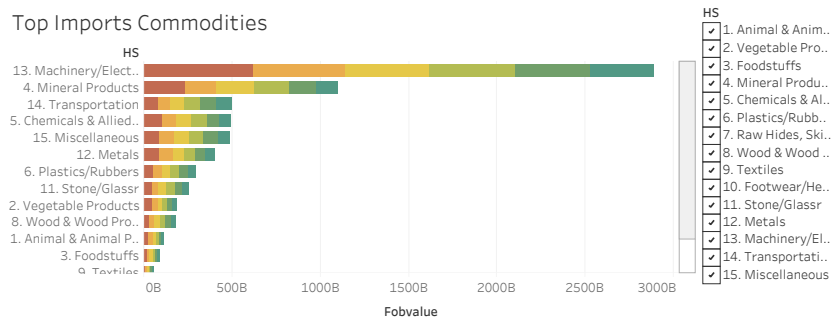
Tcosts: Total transportation/logistics costs associated with the trade.

QtyKg:Total weight of goods traded (kg).
Transport: Transport type name (e.g., Air, Sea, Road).

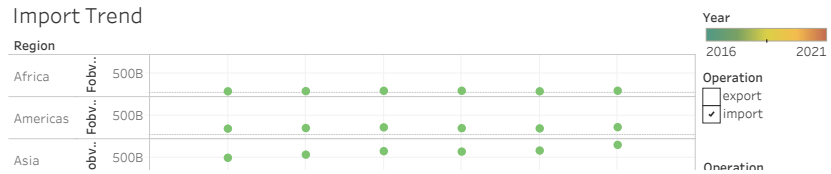
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Project overview	Data Info	Top imports	Transport Imports	Policy analysis
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Top Imports Commodities

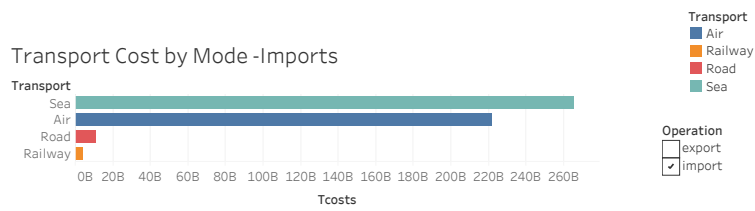


Import Trend



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Data Info	Top imports	Transport Imports	Policy analysis	Questions
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[illegible]

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Top imports	Transport Imports	Policy analysis	Questions	Top Products Sea Dependency: The highest maritime dependency appears in Mineral Products, followed by Plastics/Rubbers, Foodstuffs, Wood & Wood Products, and Vegetable Products.
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POLICY ANALYSIS

Since a large share of China's imports are transported by sea, there is a high dependency on maritime transport. Because of this strong reliance, it is important to understand which products and which supplier countries depend most on sea transport.

To analyze this, we use the Maritime Dependency Ratio (MDR), which measures the proportion of imports transported by sea compared to total imports

Calculation Fields:

MDR_Value_Product:
(FIXED [Operation], [Prod]:SUM(IF [Transport] = "Sea" THEN [Fobvalue] ELSE 0 END) / SUM([Fobvalue]))

MDR_Value_Country:
(FIXED [Operation], [Area]:SUM(IF [Transport] = "Sea" THEN [Fobvalue] ELSE 0 END) / SUM([Fobvalue]))

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Transport Imports	Policy analysis	Questions	Top Products Sea Dependency: The highest maritime dependency appears in Mineral Products, followed by Plastics/Rubbers, Foodstuffs, Wood & Wood Products, and Vegetable Products.	Top Supplier Sea Risk - The map indicates that Russia, Middle Eastern energy exporters, African resource suppliers, and South American commodity exporters show the highest maritime dependency, highlighting po...
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POLICY PERSPECTIVE QUESTIONS

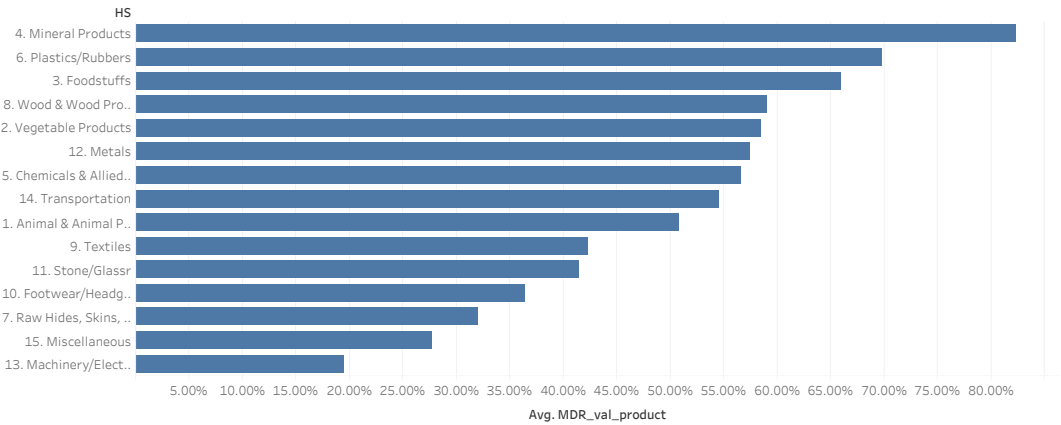
1. Which import product categories show the highest dependency on maritime transport for China?
2. Which supplier countries contribute the most to China’s sea-dependent imports?



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Policy analysis	Questions	Top Products Sea Dependency: The highest maritime dependency appears in Mineral Products, followed by Plastics/Rubbers, Foodstuffs, Wood & Wood Products, and Vegetable Products.	Top Supplier Sea Risk - The map indicates that Russia, Middle Eastern energy exporters, African resource suppliers, and South American commodity exporters show the highest maritime dependency, highlighting potential supply risks if global sea routes are disrupted.	Logistics analysis
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Top HS Sea Dependency Trends



- HS
- ☒

1. Animal & Animal Products
- ☒

2. Vegetable Products
- ☒

3. Foodstuffs
- ☒

4. Mineral Products
- ☒

5. Chemicals & Allied Industries
- ☒

6. Plastics/Rubbers
- ☒

7. Raw Hides, Skins, Leather & F..
- ☒

8. Wood & Wood Products
- ☒

9. Textiles
- ☒

10. Footwear/Headgear
- ☒

11. Stone/Glassr
- ☒

12. Metals
- ☒

13. Machinery/Electrical
- ☐

14. Transportation

Operation

☐

export

☒

import

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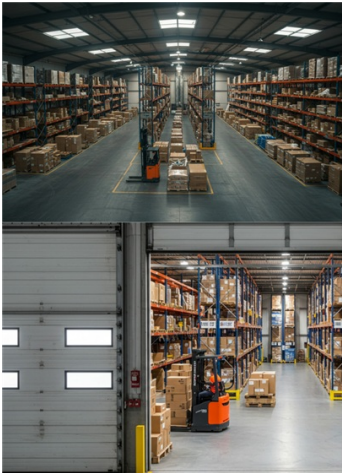
Questions	Top Products Sea Dependency: The highest maritime dependency appears in Mineral Products, followed by Plastics/Rubbers, Foodstuffs, Wood & Wood Products, and Vegetable Products.	Top Supplier Sea Risk - The map indicates that Russia, Middle Eastern energy exporters, African resource suppliers, and South American commodity exporters show the highest maritime dependency, highlighting potential supply risks if global sea routes are disrupted.	Logistics analysis	HHI
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Top Supplier Sea Risk



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LOGISTICS ANALYSIS

Herfindahl-Hirschman Index
Low Risk (HHI < 1500):* Import is spread across many suppliers – no single country dominates.

Moderate Risk (HHI 1500–2500):* A few suppliers hold significant market share – diversification is advisable.

High Risk (HHI > 2500):* Import is heavily concentrated among a limited number of suppliers – supply chain is vulnerable to disruption.

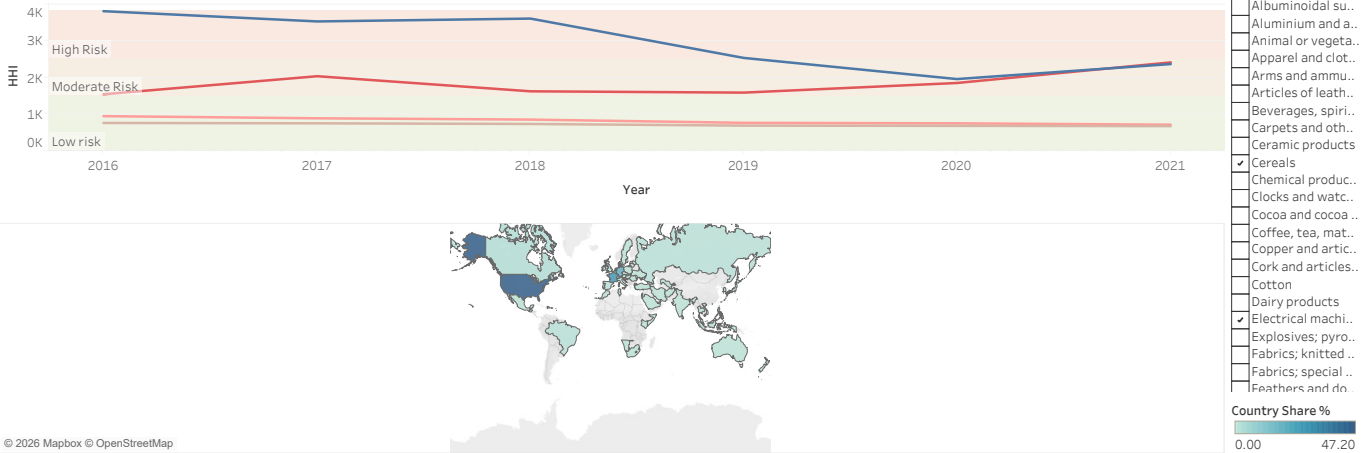
Transport Efficiency
- For the most import-dependent product categories, which alternative suppliers offer diversification potential – and what is the optimal transport mode to serve them cost-effectively?

Potential Savings
- How much can China save in EUR by switching to the optimal transport mode for its most critical imports?
- Which product-country combinations are currently overpaying for logistics relative to the cheapest available alternative – and where should the modal shift be prioritized first?

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Top Supplier Sea Risk - The map indicates that Russia, Middle Eastern energy exporters, African resource suppliers, and South American commodity exporters show the highest maritime depend..	Logistics analysis	HHI	Transport Efficiency	Recommendations
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Herfindahl-Hirschman Index



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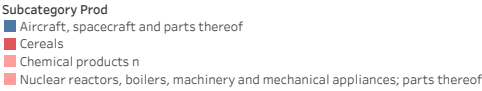
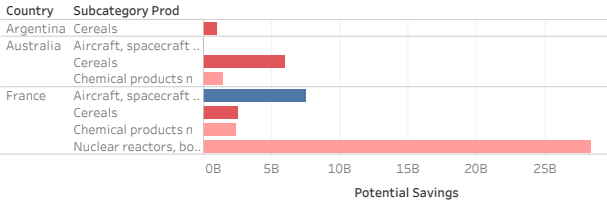
Logistics analysis	HHI	Transport Efficiency	Recommendations	Reference
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Transport efficiency

Country	Subcategory Prod	Transport			
		Air	Railway	Road	Sea
Argentina	Aircraft, spacecraft and parts thereof	7.3			4.8
	Cereals	177.0			8.7
	Chemical products n	94.9			7.3
	Nuclear reactors, boilers, machinery a..	18.4			15.1
Australia	Aircraft, spacecraft and parts thereof	8.9			6.6
	Cereals	177.5			9.6
	Chemical products n	41.5			6.5



Potential Savings



Subcategory Prod

- ☒ Aircraft, spacecr..
- ☐ Albuminoidal su..
- ☐ Aluminium and a..
- ☐ Animal or vegeta..
- ☐ Apparel and clot..
- ☐ Arms and ammu..
- ☐ Articles of leath..
- ☐ Beverages, spiri..
- ☐ Carpets and oth..
- ☐ Ceramic products
- ☒ Cereals
- ☒ Chemical produc..
- ☐ Clocks and watc..

Country

- ☐ Afghanistan
- ☐ Albania
- ☐ Algeria
- ☐ Andorra
- ☐ Angola
- ☐ Antigua and Bar..
- ☒ Argentina
- ☐ Armenia
- ☐ Aruba
- ☒ Australia
- ☒ Austria
- ☐ Azerbaijan
- ☐ Bahamas

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Logistics analysis	HHI	Transport Efficiency	Recommendations	Reference



RECOMMENDATIONS

- **Monitor high-risk categories:** Focus on imports with high sea dependency such as Mineral Products, Plastics/Rubbers, and Foodstuffs, as disruptions in maritime routes could impact supply chains.
- **Diversify suppliers:** Reduce reliance on a limited number of countries by expanding sourcing across multiple regions.
- **Develop alternative transport routes:** Encourage rail and multimodal transport options for critical imports to reduce overdependence on maritime transport.
- **Strengthen maritime resilience:** Improve port infrastructure and logistics planning to manage potential shipping disruptions.
- **Optimizing Transport Modes for Critical Imports:** Review and optimize transport modes for critical imported goods sourced from a limited number of suppliers, prioritizing cost-efficient alternatives to reduce logistics spend and lower supply chain risk.

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Logistics analysis	HHI	Transport Efficiency	Recommendations	Reference
Reference : https://unctadstat.unctad.org/datacentre/dataviewer/US.TransportCosts				